

Weather Classification Analysis Report

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1 Introduction

Dataset Overview

This analysis utilizes the `weather_classification_data.csv` dataset containing meteorological measurements for weather type prediction. The dataset comprises 13,200 samples with 15 features including temperature, humidity, wind speed, precipitation, atmospheric pressure, UV index, visibility, and weather type classifications.

Objectives

The primary objectives of this analysis are to:

- Perform comprehensive exploratory data analysis on weather data
- Build and evaluate machine learning models for weather type classification
- Compare multiple classification algorithms to identify the optimal approach
- Provide insights into meteorological patterns and feature relationships

Target Variable

The analysis focuses on predicting weather types categorized as: **Rainy, Sunny, Cloudy, and Snowy** based on available meteorological features.

2 Data Cleaning and Preparation

Dataset Structure

- Total Samples: 13,200 records
- Features: 15 columns (14 predictive features + 1 target variable)
- Target Classes: 4 weather types (3,300 samples each - perfectly balanced)
- Data Types: Mix of numerical meteorological measurements and categorical classifications

Data Quality Assessment

- Missing Values: None detected
- Data Consistency: All numerical features properly formatted
- Class Balance: Perfectly balanced (25% each)

Feature Engineering

- Categorical Encoding: Applied Label Encoding to Weather_Type (0: Rainy, 1: Sunny, 2: Cloudy, 3: Snowy)
- Feature Standardization: StandardScaler applied to all numerical features
- Feature Selection: Retained all features for comprehensive analysis

Preprocessing Results

- Standardization successful: mean ≈ 0 , std ≈ 1
- Normal-like distributions achieved for most features
- Outlier Handling: Atmospheric Pressure contained significant outliers but standardized successfully

3 Data Analysis and Visualization

Distribution Analysis

- Temperature, Humidity, Wind Speed: Normal distributions
- Atmospheric Pressure: Widest range with outliers
- UV Index & Precipitation: Slight skewness, acceptable standardization

Correlation Analysis

Strong Positive Correlations:

- Humidity $\leftrightarrow$ Precipitation: 0.64
- Wind Speed $\leftrightarrow$ Precipitation: 0.44
- Temperature $\leftrightarrow$ UV Index: 0.37

Notable Negative Correlations:

- Humidity $\leftrightarrow$ Visibility: -0.48
- Precipitation $\leftrightarrow$ Visibility: -0.46

Target Variable Correlation:

- UV Index: 0.35 (highest individual correlation)
- Others $\leq 0.25 \rightarrow$ multi-feature approach required

Feature-Target Relationship

- Best Predictors: UV Index, Precipitation, Visibility
- Moderate Predictors: Humidity, Atmospheric Pressure
- Weakest Predictor: Wind Speed

Class Separability: Distinct patterns confirm machine learning classification feasibility.

4 Model Building

Model Selection

Three algorithms selected:

1. **Logistic Regression** Baseline linear classifier. Configuration: multi-class one-vs-rest, max_iter=1000. Strengths: simple, interpretable.
2. **Decision Tree** Captures non-linear feature interactions. Configuration: max_depth=10, min_samples_split=20.
3. **Random Forest** Ensemble model for robustness & reduced overfitting. Configuration: n_estimators=100, max_depth=15, min_samples_split=10.

Training Process

- Data Split: 80% training (10,560), 20% testing (2,640)
- Stratification: Preserved class distribution
- Cross-validation: Stratified sampling for fairness

5 Model Evaluation

Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	91.25%	91.37%	91.25%	91.28%
Decision Tree	90.87%	90.89%	90.87%	90.88%
Logistic Regression	84.89%	85.11%	84.89%	84.90%

Analysis

- Best Performer: Random Forest (91.25%)
- Tree-based models outperform Logistic Regression (~6% higher)
- All models balanced in precision & recall → robust classification

Interpretation

- Confirms EDA insights about complex feature interactions
- Validates UV Index & Precipitation importance
- Ensemble methods reduce variance, improving reliability

6 Conclusion

Summary of Findings

- Dataset: Clean, balanced, well-prepared
- Feature Relationships: Logical, meteorologically valid
- Models: All >84% accuracy, Random Forest optimal

Key Insights

- Weather classification requires multi-feature inputs
- UV Index & Precipitation strongest predictors
- Ensemble methods best suited for this dataset

Recommendations

- Deploy Random Forest for operational predictions
- Explore advanced ensembles (XGBoost, LightGBM)
- Implement monitoring for real-world applications

Future Work

- Feature engineering for correlated variables
- Deep learning for complex pattern recognition
- Time series modeling with timestamp data
- Real-time prediction pipelines

Final Assessment

Weather type classification is achievable with **91.25% accuracy** using ensemble ML methods. Random Forest provides a robust foundation for practical weather prediction applications.